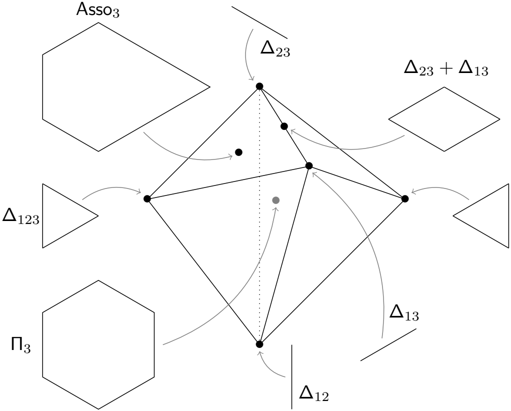
Deformed permutahedra
An inductive construction of the submodular cone

Germain Poullot with Georg Loho & Arnau Padrol



14 January 2026, Bielefeld arXiv:2510.03177

1. Deformations (a.k.a. Minkowski summands)
    - Deformations
    - Cone of deformations
2. Deformed permutahedra = submodularity
    - Combinatorics of the permutahedron
    - Submodular functions
3. Submodular Cone in general
    - Known facts about $\mathbb{SC}_n$
    - Submodular cone $n = 4$ (and $n = 5$)
4. Inductive construction of (rays of) $\mathbb{SC}_n$
    - GP-sum
    - Rays of $\mathbb{SC}_n$

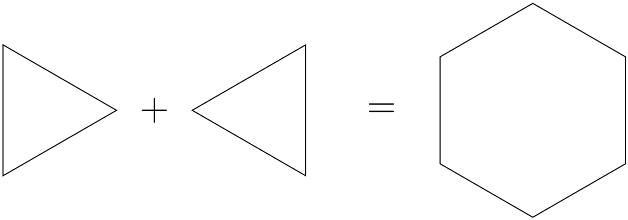
Deformations (a.k.a. Minkowski summands)

Definition

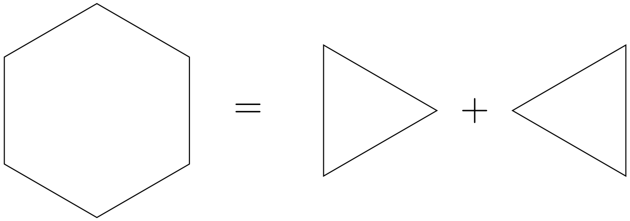
P, Q polytopes. *Minkowski sum*:

$$P + Q = \{\boldsymbol{p} + \boldsymbol{q} \ ; \ \boldsymbol{p} \in P, \ \boldsymbol{q} \in Q\}$$

N.B. $\mathrm{Vert}(P+Q) \subseteq \mathrm{Vert}(P) + \mathrm{Vert}(Q)$



Minkowski sum



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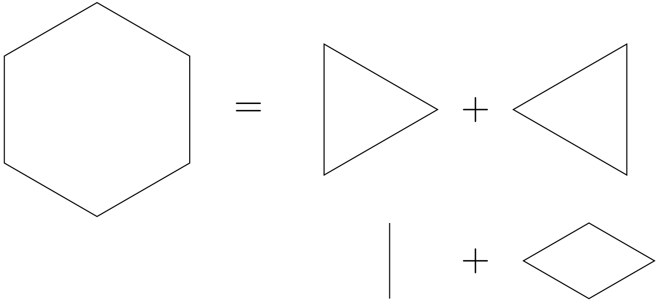

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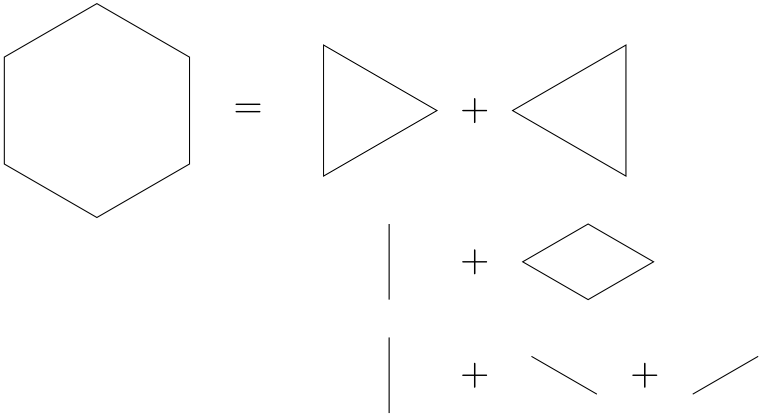

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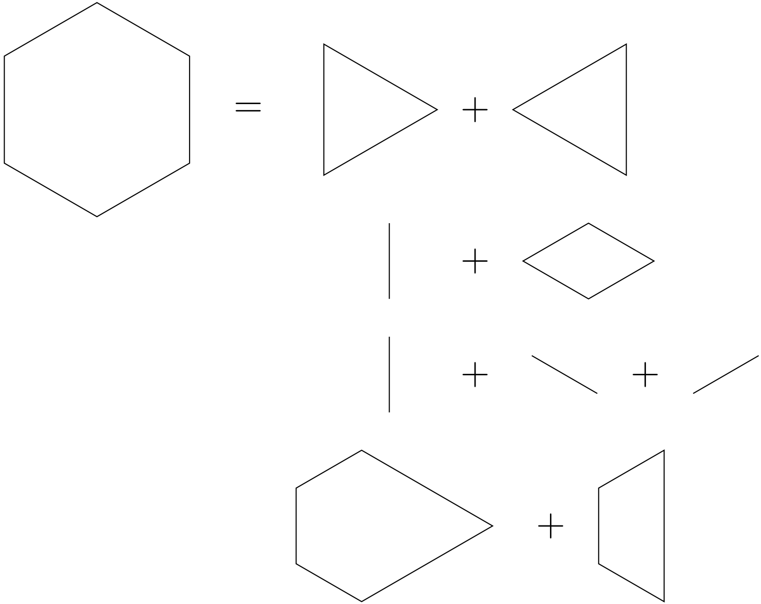

Minkowski summands

Definition

Q is a *Minkowski summand*, a.k.a. *deformation*, of P when there exists R and $\lambda > 0$ with:

$$\lambda P = Q + R$$

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What is the best way to write P as a Minkowski sum ?

- With the fewest number of (indecomposable) summands ?
- With the (indecomposable) summands of smallest dimension ?
- Respecting some symmetries ? ...

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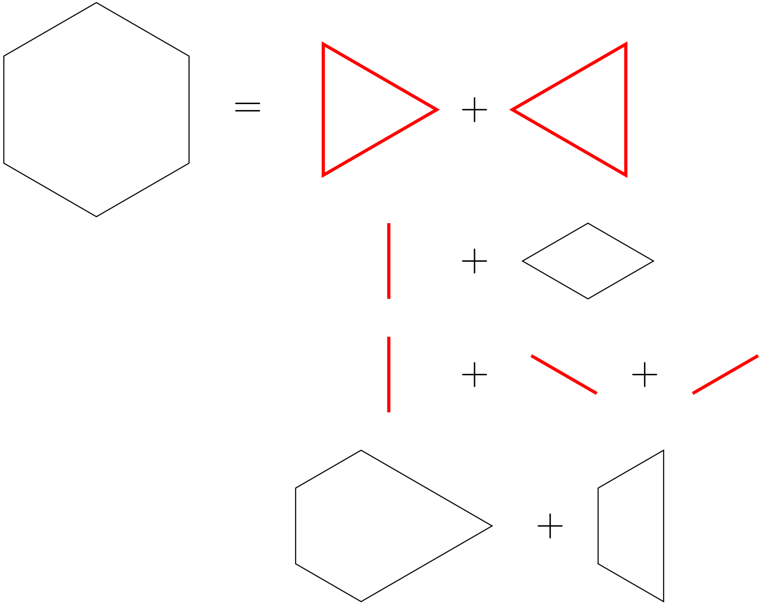
$\Longrightarrow$ What is the structure of $\mathbb{DC}(\mathrm{P})$?

For indecomposable, see *Indecomposability and beyond via the graph of edge dependencies*, arXiv:2512.05307, with Arnau Padrol

Minkowski summands

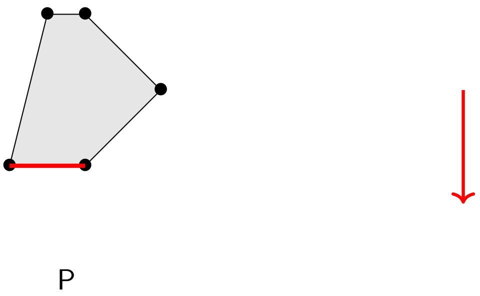


Definition

Polytope: convex hull of finitely many points in $\mathbb{R}^n$
bounded intersection of finitly many half-spaces in $\mathbb{R}^n$

Definition

Face: $\mathrm{P}^{\boldsymbol{c}} := \left\{\boldsymbol{x} \in \mathbb{R}^n \ ; \ \langle \boldsymbol{x}, \boldsymbol{c} \rangle = \max_{\boldsymbol{y} \in \mathrm{P}} \langle \boldsymbol{y}, \boldsymbol{c} \rangle \right\}$



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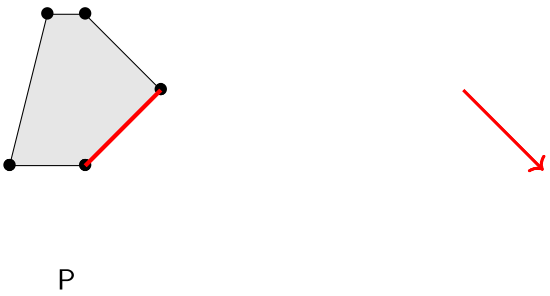

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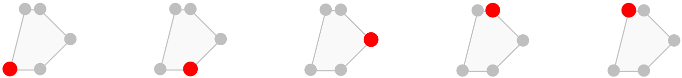
P

Face lattice

Face lattice: poset of inclusions of faces



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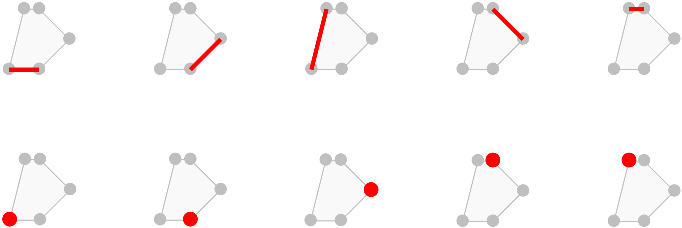

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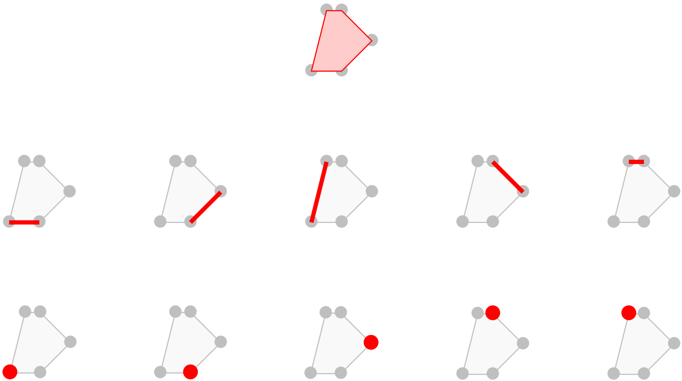

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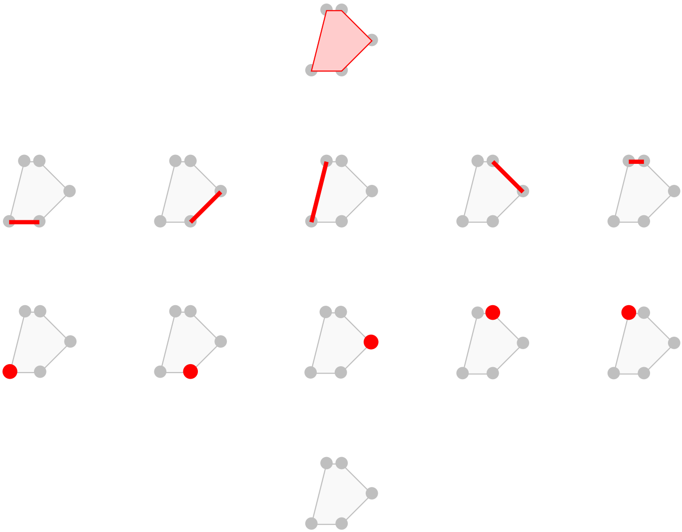

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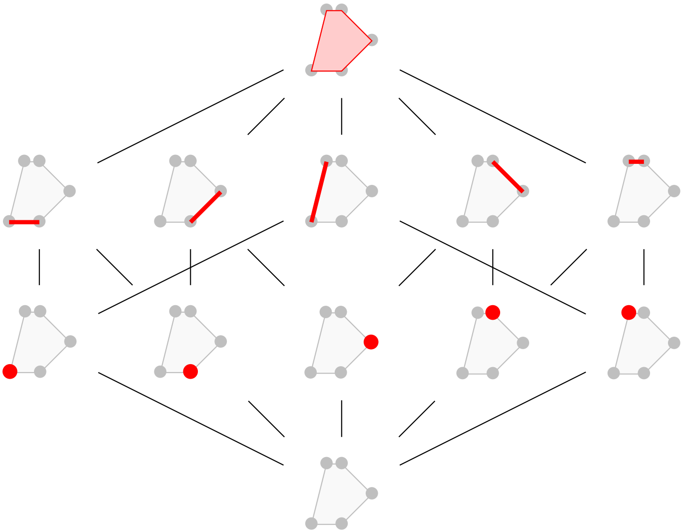
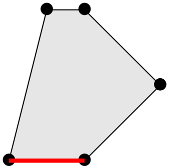


Definition

Normal cone of a face F: $\mathcal{N}_{\mathsf{P}}(\mathsf{F}) := \{\boldsymbol{c} \; ; \; \mathsf{P}^{\boldsymbol{c}} = \mathsf{F}\}$
Normal fan $\mathcal{N}_{\mathsf{P}}$: collection of $\mathcal{N}_{\mathsf{P}}(\mathsf{F})$ for F face of P



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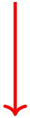

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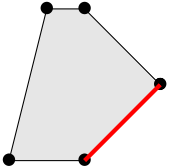

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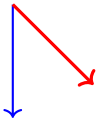

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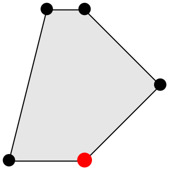

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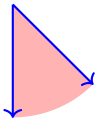

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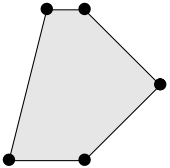

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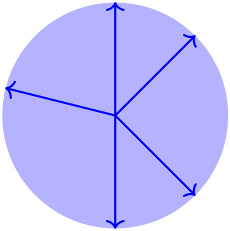

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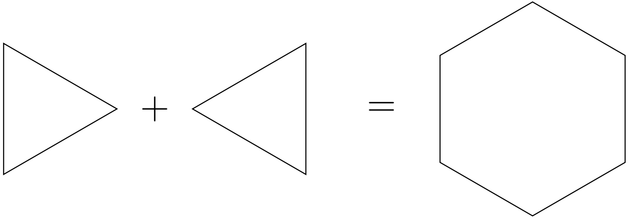
Minkowski sum seen with normal fans

Lemma

$\mathcal{N}_{\mathsf{P+Q}} =$ *common refinement of* $\mathcal{N}_{\mathsf{P}}$ *and* $\mathcal{N}_{\mathsf{Q}}$



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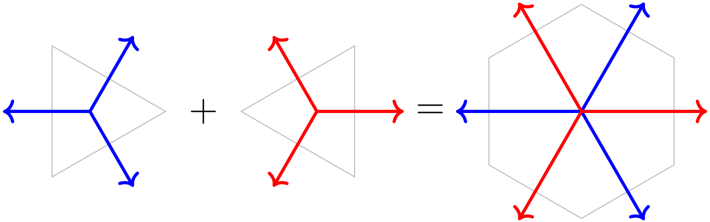
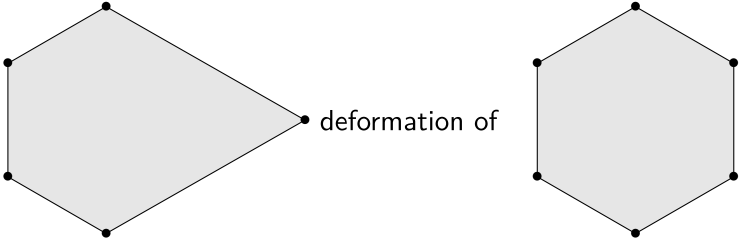


$\mathcal{N}_{\mathsf{P}}$: Normal fan of polytope P

Coarsening: Choose maximal cones and merge them

Theorem

Q *is a deformation of* P *iff* $\mathcal{N}_{\mathsf{Q}}$ *coarsens* $\mathcal{N}_{\mathsf{P}}$.



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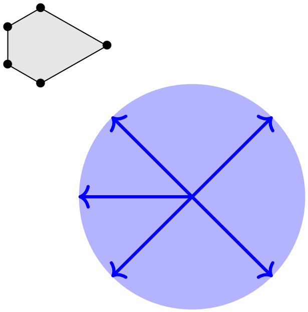

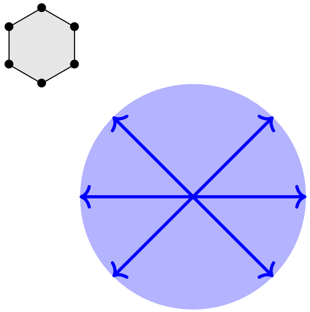

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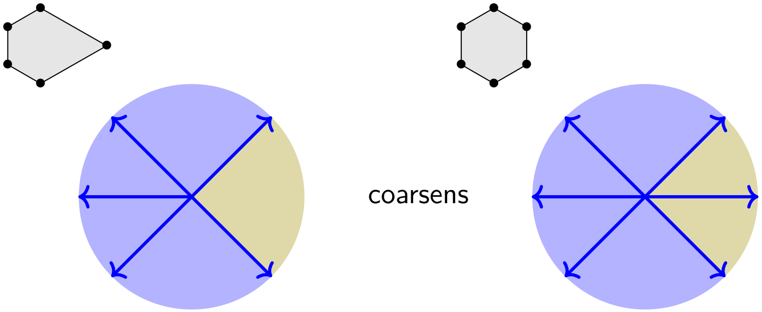
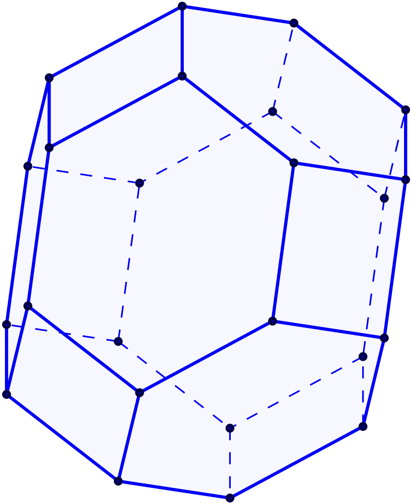
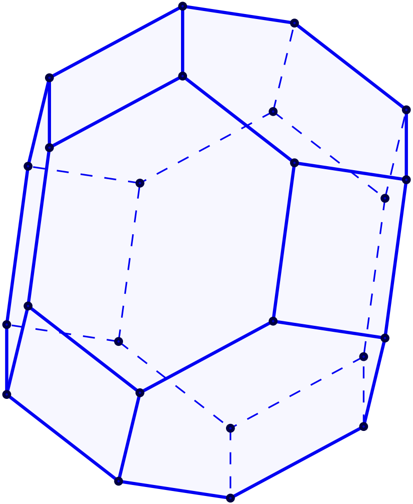


Deformations of 3-dim permutahedron



Permutahedron Π_4



Sequence of deformations of Π_4

Height deformation cone

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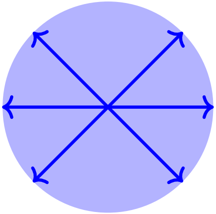

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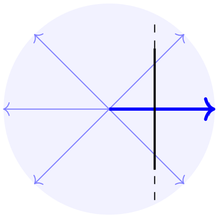

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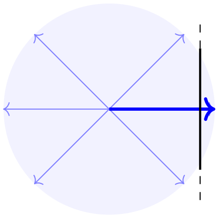

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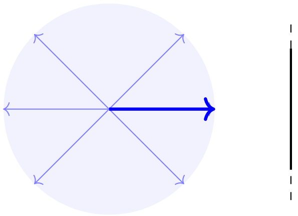

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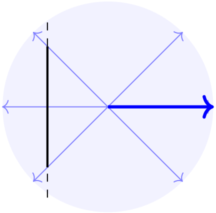

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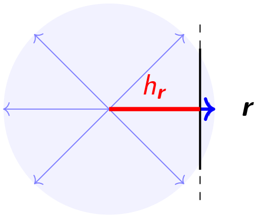

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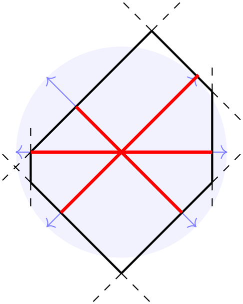

Parametrization:

height vector:
$\boldsymbol{h} = \big(h_{\boldsymbol{r}}\big)_{\boldsymbol{r} \text{ rays}}$

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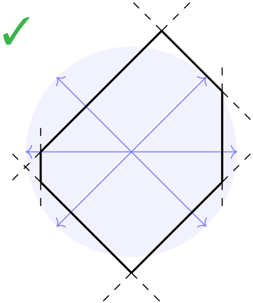

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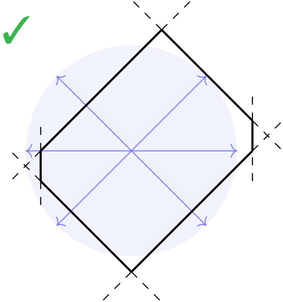

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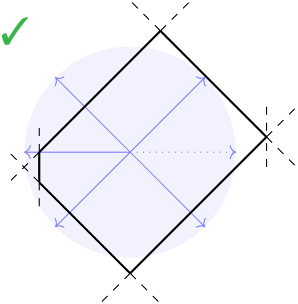

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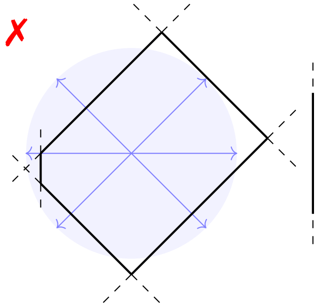

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Wall-crossing inequalities:

linear inequalities on $\boldsymbol{h}$

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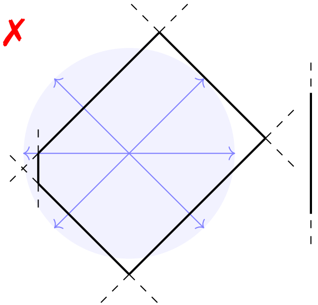

<u>Parametrization:</u>

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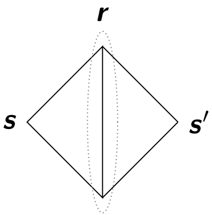
Wall-crossing inequalities:
linear inequalities on $\boldsymbol{h}$

$$P_{\boldsymbol{h}} = \{\boldsymbol{x} \; ; \; \langle \boldsymbol{x} \,|\, \boldsymbol{r} \rangle \leq h_{\boldsymbol{r}}\}$$

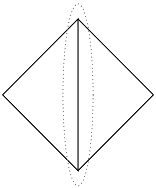
Wall-crossing inequalities



If, with $\alpha_{\boldsymbol{s}}, \alpha_{\boldsymbol{s}'} > 0$

$$\alpha_{\boldsymbol{s}} \boldsymbol{s} + \alpha_{\boldsymbol{s}'} \boldsymbol{s}' + \sum_{\boldsymbol{r}} \alpha_{\boldsymbol{r}} \boldsymbol{r} = \boldsymbol{0}$$

Wall-crossing inequalities



If, with $\alpha_{\boldsymbol{s}}, \alpha_{\boldsymbol{s}'} > 0$

$$\alpha_{\boldsymbol{s}} \boldsymbol{s} + \alpha_{\boldsymbol{s}'} \boldsymbol{s}' + \sum_{\boldsymbol{r}} \alpha_{\boldsymbol{r}} \boldsymbol{r} = \boldsymbol{0}$$

then

$$\alpha_{\boldsymbol{s}} h_{\boldsymbol{s}} + \alpha_{\boldsymbol{s}'} h_{\boldsymbol{s}'} + \sum_{\boldsymbol{r}} \alpha_{\boldsymbol{r}} h_{\boldsymbol{r}} \geq 0$$

Edge-length deformation cone

Theorem

Q *deformation of* P $\Leftrightarrow$ *same edge-directions, but different lengths*

Definition

Edge-length deformation cone: $\mathbb{DC}(\mathrm{P}) = \{\mathrm{Q} \ ; \ \mathrm{Q} \text{ same edge-dir } \mathrm{P}\}$

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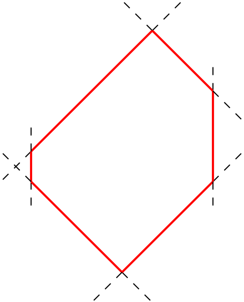

Parametrization:

edge-length vector:

$\boldsymbol{\ell} = \left(\ell_{\boldsymbol{e}}\right)_{\boldsymbol{e} \text{ edge}}$

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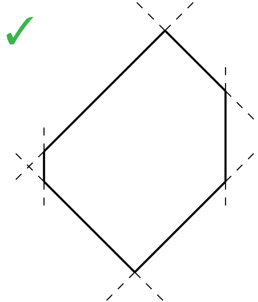

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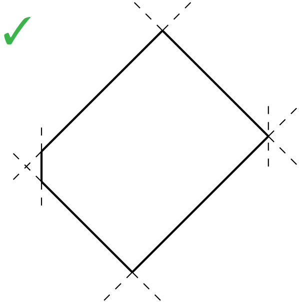

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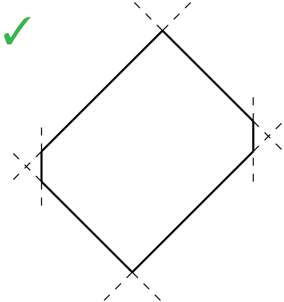

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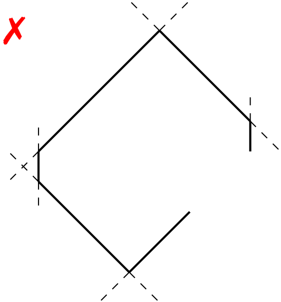

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Polygonal face equations:

linear equations on $\boldsymbol{\ell}$

$\ell_{\boldsymbol{e}} \geq 0$ for all $\boldsymbol{e}$ edge

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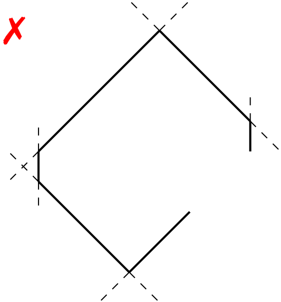

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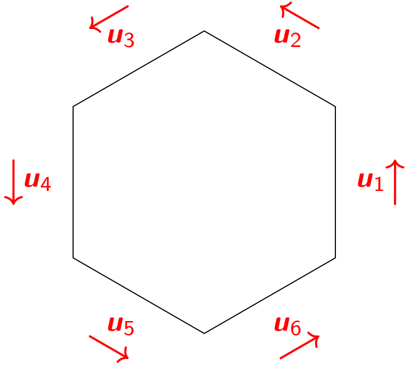
$\mathrm{P}_{\boldsymbol{\ell}} =$ start at a vertex, find the coordinates of the other vertices from the graph of P and $\boldsymbol{\ell}$

Polygonal face equations



For F a 2-dim face of P:

$$\sum_{\boldsymbol{e}} \boldsymbol{u}_{\boldsymbol{e}} = \mathbf{0} \quad , \quad \boldsymbol{u}_{\boldsymbol{e}} \text{ unit vector}$$

hence

$$\sum_{\boldsymbol{e}} \ell_{\boldsymbol{e}} \boldsymbol{u}_{\boldsymbol{e}} = \mathbf{0}$$

Summary on $\mathbb{DC}(\mathrm{P})$

$\mathbb{DC}(\mathrm{P})$

Q	$\boldsymbol{\ell}$	$\boldsymbol{h}$
Minkowski summands	edge-lengths	heights on rays
$\mathrm{Q}_1 + \mathrm{Q}_2$	$\boldsymbol{\ell}_1 + \boldsymbol{\ell}_2$	$\boldsymbol{h}_1 + \boldsymbol{h}_2$
Dilation $\lambda\mathrm{Q}$	$\lambda\boldsymbol{\ell}$	$\lambda\boldsymbol{h}$
Translations	None	Lineal
	edge directions	normal fan $\mathcal{N}_\mathrm{P}$
complicated	Polygonal face eq.	Wall-crossing ineq.
	V-description	H-description
Polytope algebra	Weight algebra	Polynomial algebra

$\mathbb{DC}(\mathrm{P})$ is a ray = P Minkowski indecomposable

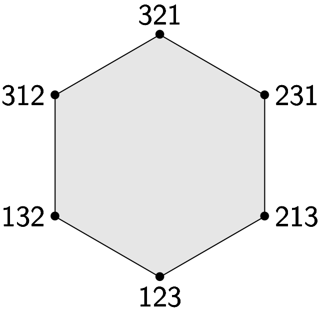
$\mathbb{DC}(\mathrm{P})$ is simplicial cone = P has **unique** Minkowski decomposition

Deformed permutahedra = submodularity

Permutahedron

Example (Permutahedron)

$$\Pi_n = \operatorname{conv}\left\{ \begin{pmatrix} \sigma(1) \\ \vdots \\ \sigma(n) \end{pmatrix} \ ; \ \sigma \text{ permutation of } \{1, \dots, n\} \right\}$$



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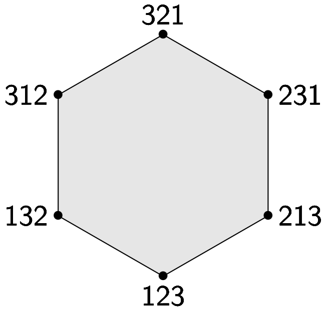

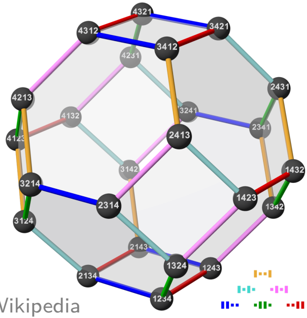

Wikipedia

Braid fan

Definition

Braid fan: arrangement of hyperplanes $H_{i,j} := \{\boldsymbol{x} \;;\; x_i = x_j\}$

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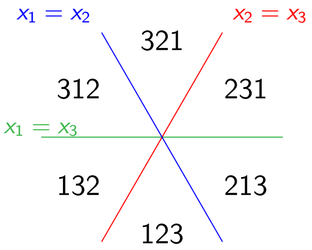
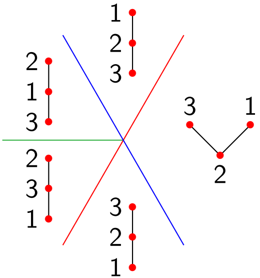
Definition

Generalized permutahedron: deformation of Π_n

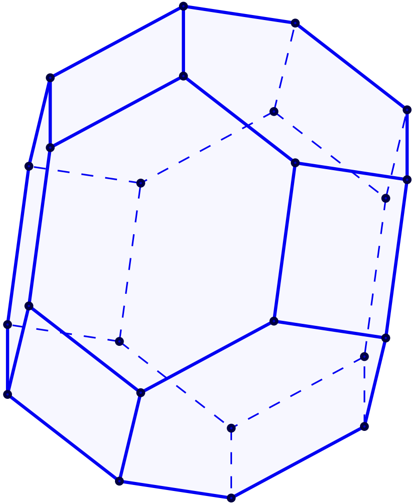
i.e. P generalized permutahedron iff $\mathcal{N}_{\mathrm{P}}$ coarsens braid fan

i.e. P generalized permutatahedron iff edges in directions $\boldsymbol{e}_i - \boldsymbol{e}_j$

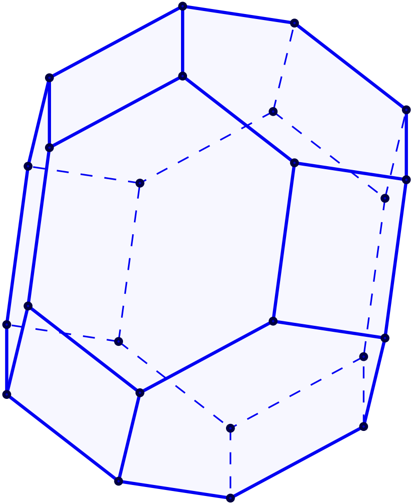




Deformations of Π_4

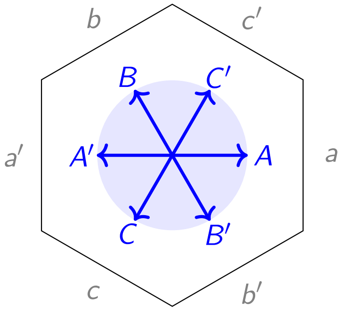


Permutahedron Π_4



Sequence of deformations of Π_4

2-dimensional example



Wall-crossing inequalities:

$h_A + h_B \geq h_{C'}$
$h_B + h_C \geq h_{A'}$
$h_C + h_A \geq h_{B'}$
& 3 others ineq.

Polygonal face equations:

$\ell_a - \ell_{a'} = \ell_b - \ell_{b'} = \ell_c - \ell_{c'}$
& $\boldsymbol{\ell} \in \mathbb{R}^6_+$

Submodular functions

Definition

Submodular functions $\boldsymbol{h} : 2^{[n]} \to \mathbb{R}$

$$\forall A, B \subseteq [n],\ \ h(A) + h(B) \geq h(A \cap B) + h(A \cup B)$$

In bijection with generalized permutahedra:
$\mathrm{P}_{\boldsymbol{h}} = \{\boldsymbol{x}\ ;\ \sum_{i \in A} x_i \leq h(A)$ for all $A \subseteq [n]$ and $\sum_{i=1}^{n} x_i = h([n])\}$

Submodular Cone in general

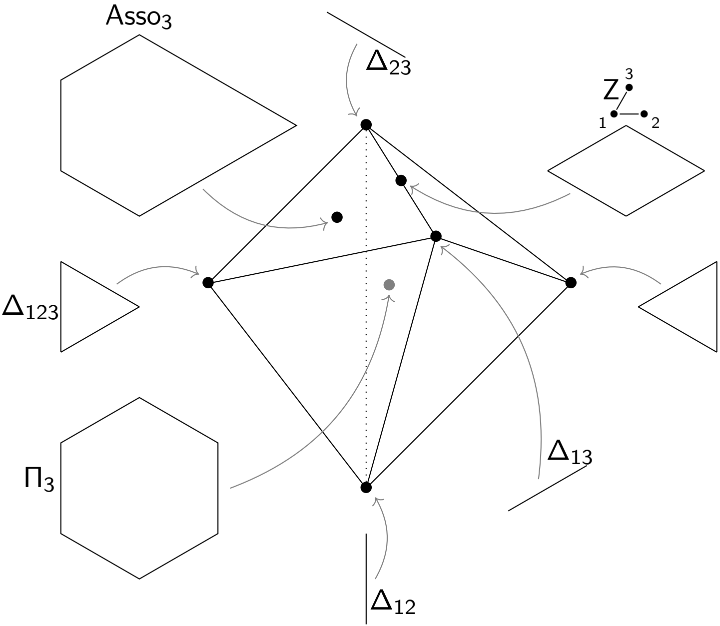
Submodular Cone

Definition

Submodular cone: deformation cone of the permutahedron Π_n

	$\mathbb{SC}_n$
Dim (no lineal)	$2^n - n - 1$
# facets	$\binom{n}{2}2^{n-2}$
# rays	unknown!

Submodular Cone for Π_3



Submodular Cone's faces

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But that belongs to a different presentation...

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	$\mathbb{SC}_n$	$\mathbb{DC}(\mathrm{Asso}_n)$
Dim (no lineal)	$2^n - n - 1$	$\binom{n}{2}$
# facets	$\binom{n}{2}2^{n-2}$	$\binom{n}{2}$
# rays	unknown!	$\binom{n}{2}$
		simplicial!

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	$\mathbb{SC}_n$	$\mathbb{DC}(\mathrm{Asso}_n)$	Graph zono.	Nestohedra
Dim (no lineal)	$2^n - n - 1$	$\binom{n}{2}$	✓	✓
# facets	$\binom{n}{2}2^{n-2}$	$\binom{n}{2}$	✓	✓
# rays	unknown!	$\binom{n}{2}$	✗	✗
		simplicial!		

But that belongs to a different presentation...

Recall: dim $= 11$, nbr facets $= 80$

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Draw all generalized permutahedra ? (ask computer)

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22 107 faces (Please do not draw...)

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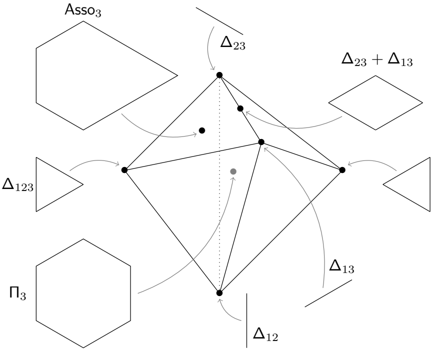
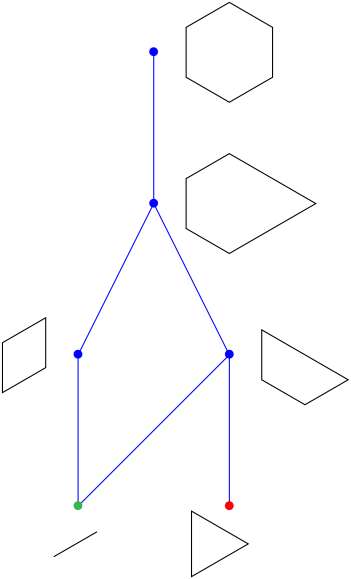
$\Longrightarrow$ quotient by symmetries

Symmetries of the braid fan

Braid symmetries: permutation of coordinates + central symmetry





Drawing $\mathbb{SC}_4$

Recall: $\dim = 11$, nbr facets $= 80$

Draw all generalized permutahedra ? (ask computer)

$$22\ 107 \text{ faces}$$

$\Longrightarrow$ quotient by symmetries

Drawing $\mathbb{SC}_4$

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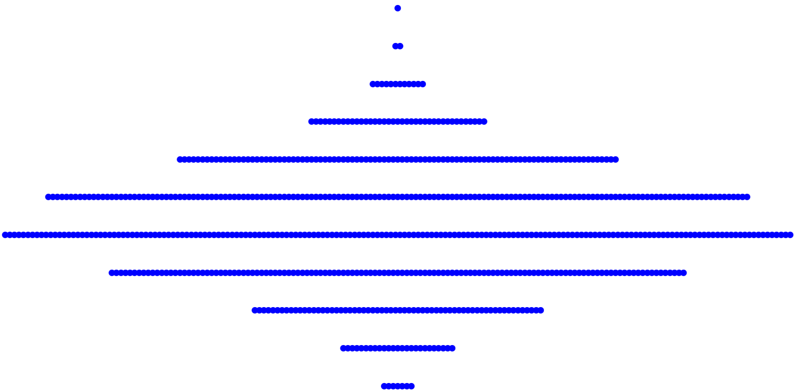
22 107 faces

$\Longrightarrow$ quotient by symmetries

703 "faces"

Reduced face lattice of $\mathbb{SC}_4$



d	1	2	3	4	5	6	7	8	9	10	11	total
f_d	7	25	64	127	174	155	97	39	12	2	1	703

Reduced f-vector of $\mathbb{SC}_n$

Reduced $\mathbb{SC}_n$ f-vector:

$n = 3$
$\dim \mathbb{SC}_3 = 4$
$(2, 2, 1, 1)$

$n = 4$
$\dim \mathbb{SC}_4 = 11$
$(7, 25, 64, 127, 174,$
$155, 97, 39, 12, 2, 1)$

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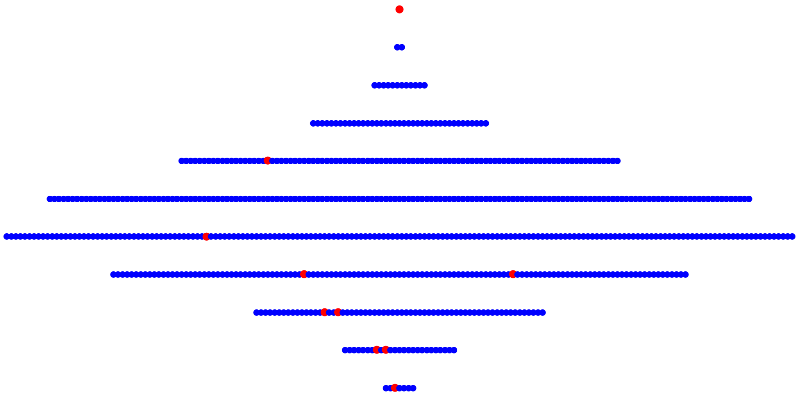
Thanks to Winfried Bruns for helping compute!

Database for dim 1-4 & 19-26

$n = 5$, dim $\mathbb{SC}_5 = 26$

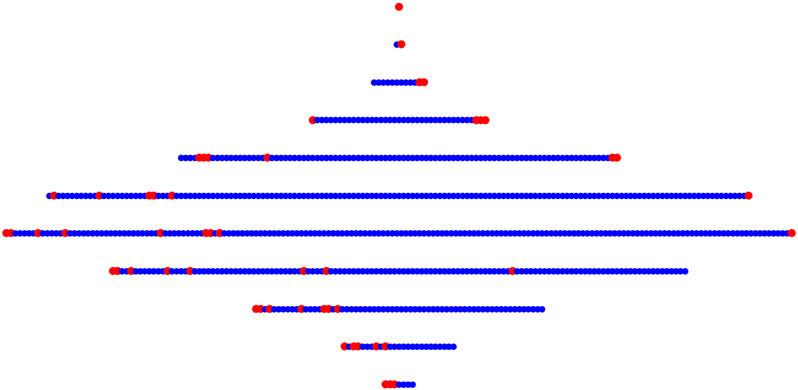
$*$672
$*$24 026
$*$373 433
$*$3 355 348
19 739 627
81 728 494
249 483 675
579 755 845
1 048 953 035
1 501 555 944
1 719 688 853
1 587 510 812
1 186 372 740
719 012 097
353 190 577
140 265 886
44 831 594
11 464 559
$*$2 326 596
$*$372 031
$*$46 330
$*$4 572
$*$355
$*$30
$*$2
$*$1

Graphical zonotopes & Nestohedra are sparse

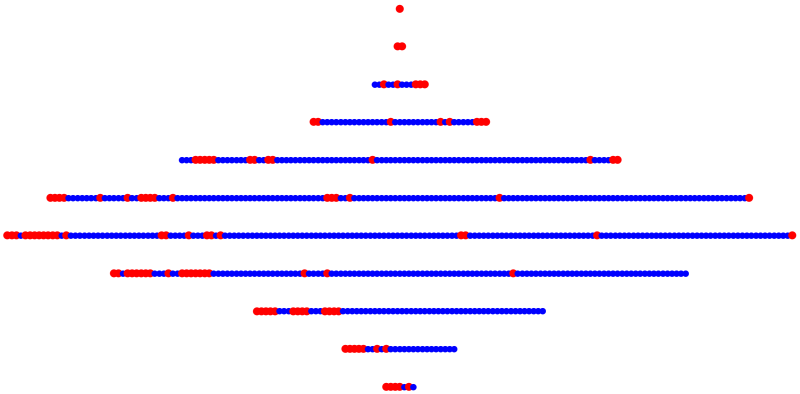


With: Graphical Zonotopes
10 polytopes

Graphical zonotopes & Nestohedra are sparse

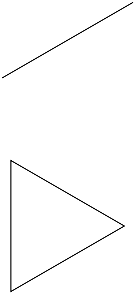


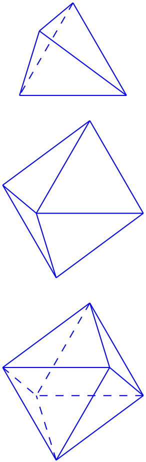
With: Graphical Zonotopes & Nestohedra
10 + 46 polytopes

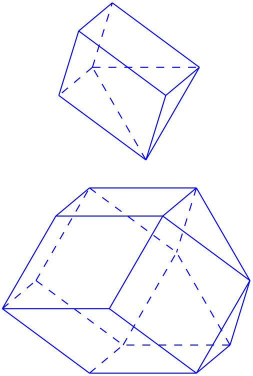


With: Graphical Zono & Nestohedra $\subsetneq$ Hypergraphic Polytopes,
$+$ Shard Polytopes, Quotientopes,
$+$ Matroid Polytopes
$=$ 112 polytope (only...)

What about the rays of $\mathbb{SC}_4$?





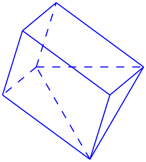


Then 665 of dim 4,
$>$ 126 629 of dim 6

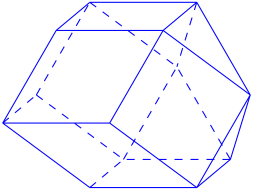
Deformed permutahedra:

Minkowski indecomposable ✔

Matroid Polytopes ✗



Strawberry



Persimmon

Deformed permutahedra:

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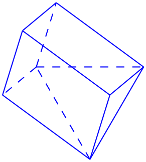
Matroid Polytopes ✗

Hypergraphic polytopes ✗

Shard Polytopes ✗



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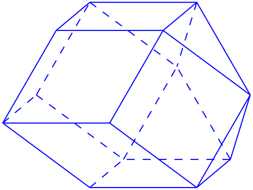

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Deformed permutahedra:

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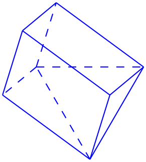
Shard Polytopes ✗

Persimmon:
Polypositroid ✗
Removahedron ✗



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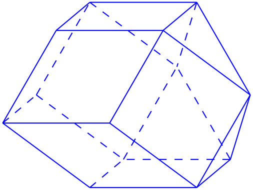

Persimmon

Number of rays of $\mathbb{SC}_n$

Theorem ((approximately) Nguyen, '86)

The number t_n of rays of $\mathbb{SC}_n$ satisfies

$$2^{2^n \cdot n^{-3/2}} \leq t_n$$

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Theorem (Loho–Padrol–P., '25+)

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$$2^{2^{n-2}} \leq t_n$$

N.B.: By Upper Bound Theorem: $t_n \leq n^{2^n}$.

Precisely: $n - 2 \leq \log_2 \log_2 t_n \leq n + \log_2 \log_2 n + 1$

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What's 2^{2^n}?

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$\Rightarrow$ I want to construct 1 rays of $\mathbb{SC}_{n+1}$ for each **pair** of rays of $\mathbb{SC}_n$

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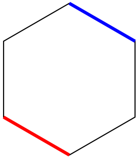
Let's do an induction!

Inductive construction of (rays of) $\mathbb{SC}_n$

P^+, P^-: **opposite** faces, maximizing/minimizing $\boldsymbol{e}_{n+1}$ on P.



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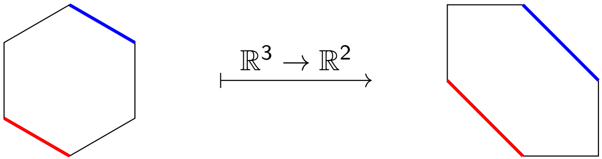

GP-sum

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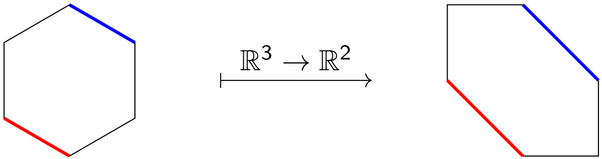

Theorem (Frank '11)

Deformed permutahedra are ***uniquely*** *determined by their pair of top and bottom faces.*
Thus, $\mathrm{P} \mapsto (\mathrm{P}^+, \mathrm{P}^-)$ *is a bijection:*

What's the reciprocal bijection?

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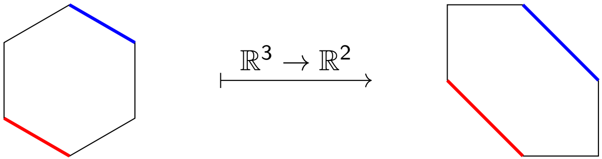

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Thus, $P \mapsto (P^+, P^-)$ *is a bijection:* $\mathbb{SC}_{n+1} \simeq$ *(roughly* $\mathbb{SC}_n^2$*)*

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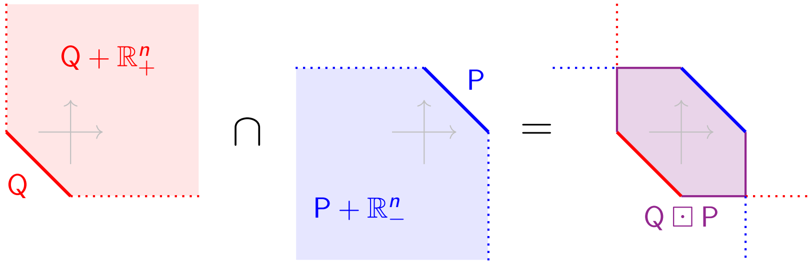

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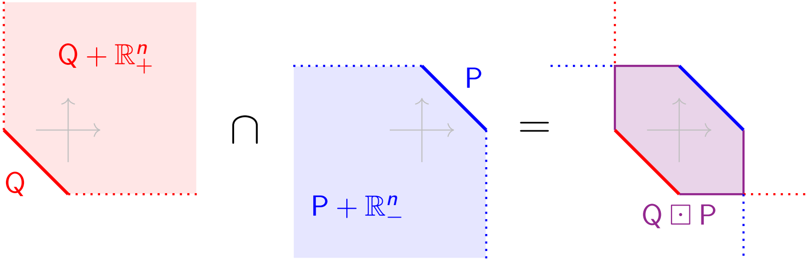
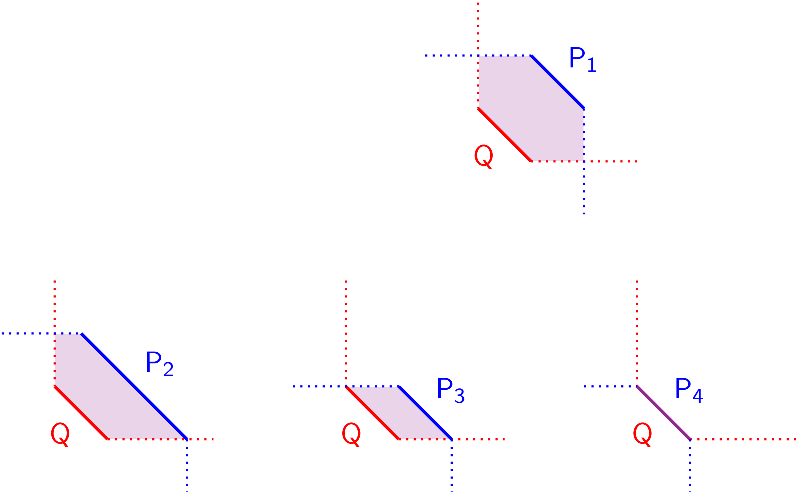


projection of $\boldsymbol{x} \in \mathbb{R}^{n+1}$ is $\downarrow(\boldsymbol{x}) := (x_1, \ldots, x_n)$
lift of $\boldsymbol{x} \in \mathbb{R}^n$ is $\uparrow(\boldsymbol{x}) := (x_1, \ldots, x_n, -\sum_{i=1}^n x_i)$

Reciprocal of $\mathrm{P} \mapsto (\downarrow(\mathrm{P}^+), \downarrow(\mathrm{P}^-))$ is the map $(\mathrm{P}, \mathrm{Q}) \mapsto \uparrow(\mathrm{Q} \boxdot \mathrm{P})$

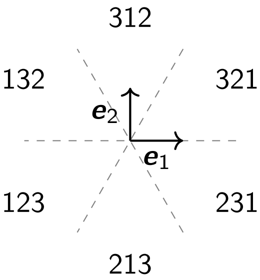
GP-sums

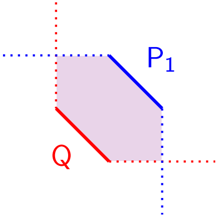
GP-sum exists iff $P \subseteq (Q + \mathbb{R}^+)$ and $Q \subseteq (P + \mathbb{R}^-)$

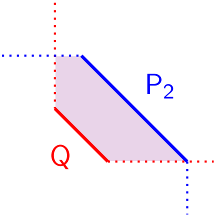


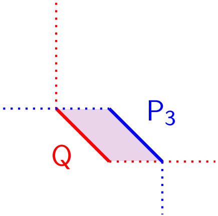
GP-sums

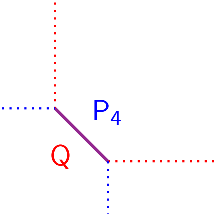
GP-sum exists ~~iff $\mathrm{P} \subseteq (\mathrm{Q} + \mathbb{R}^+)$ and $\mathrm{Q} \subseteq (\mathrm{P} + \mathbb{R}^-)$~~
iff for all $i \in [n], \sigma \in S_n$, we have $\mathrm{P}_i^\sigma \leq \mathrm{Q}_i^\sigma$











Inductive process on $\mathbb{SC}_n$

Theorem (Frank '11, rewritten)

There is a bijection between $\mathbb{SC}_{n+1}$ *and pairs* $(\mathsf{P},\mathsf{Q}) \in \mathbb{SC}_n^2$ *with* $\forall i, \sigma,\ \mathsf{P}_i^\sigma \leq \mathsf{Q}_i^\sigma$. *Namely:*

$$\begin{aligned} \mathsf{P} &\mapsto \Big(\downarrow(\mathsf{P}^+), \downarrow(\mathsf{P}^-)\Big) \\ (\mathsf{P},\mathsf{Q}) &\mapsto \uparrow(\mathsf{Q} \boxdot \mathsf{P}) \end{aligned}$$

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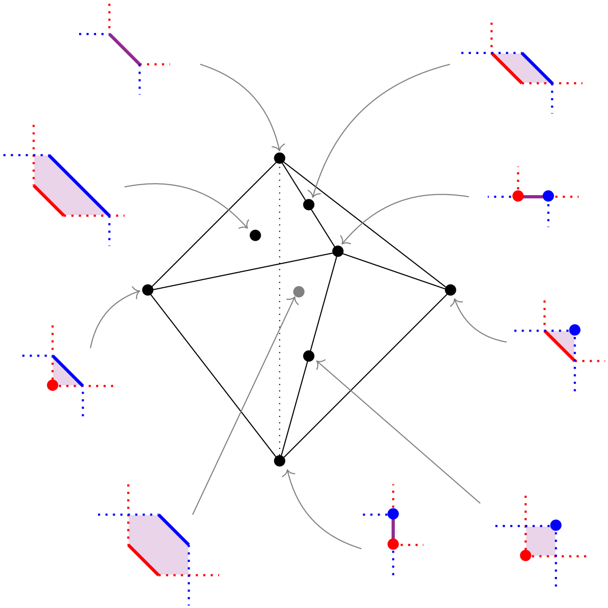
Problem: inductive process on $\mathbb{SC}_n$, not on the <u>face lattice</u> of $\mathbb{SC}_n$

Not inductive process on face lattice of $\mathbb{SC}_n$



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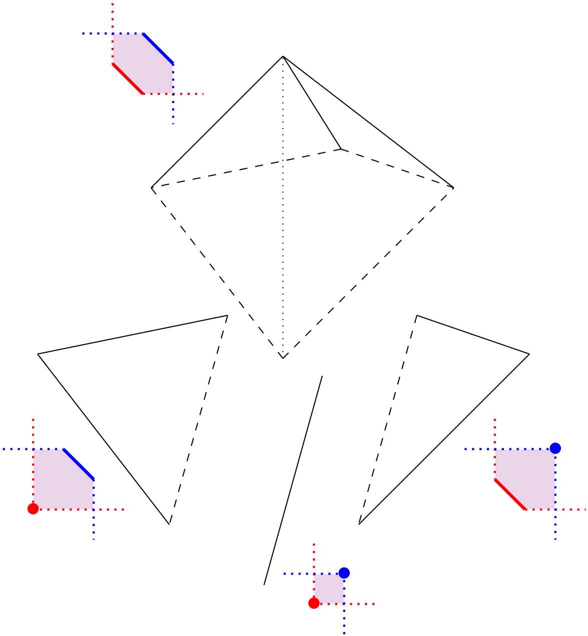

Inductive construction of rays

Problem: P and Q rays of $\mathbb{SC}_n$ $\quad\not\Rightarrow\quad$ $\uparrow(\mathrm{Q} \boxdot \mathrm{P})$ rays of $\mathbb{SC}_{n+1}$

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Theorem (Loho–Padrol–P., '25+)

P, Q *rays of* $\mathbb{SC}_n$ + *small condition*
$\Rightarrow \exists \lambda > 0, \boldsymbol{t} \in \mathbb{R}^n,\ (\lambda \mathrm{Q} + \boldsymbol{t}) \boxdot \mathrm{P}$ *ray of* $\mathbb{SC}_{n+1}$.

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Small condition: fertility
(P, Q) is *fertile* iff there exists $i \in [n]$ such that for all $\tau \in S_n$,
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Lemma

If (S, P) *is fertile, then* $\big(\uparrow(\mathrm{S} \boxdot \mathrm{R}), \uparrow(\mathrm{Q} \boxdot \mathrm{P})\big)$ *is also fertile.*

Good news: Fertility is hereditary!

Number of rays

Lemma (Consequently)

If $\{\mathrm{P}_1, \mathrm{P}_2, \ldots, \mathrm{P}_r\}$ *are rays of* $\mathbb{SC}_n$ *with* $(\mathrm{P}_i, \mathrm{P}_j)$ *fertile for all* i, j, *then there exist* $\lambda_{i,j} > 0, \boldsymbol{t}_{i,j} \in \mathbb{R}^n$ *such that* $\left\{\uparrow((\lambda_{i,j}\mathrm{P}_j + \boldsymbol{t}_{i,j}) \boxdot \mathrm{P}_i)\ ;\ 1 \leq i, j \leq r\right\}$ *are rays of* $\mathbb{SC}_{n+1}$ *also pairwise fertile.*

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I can construct 1 rays of $\mathbb{SC}_{n+1}$ for each *fertile* pair of rays of $\mathbb{SC}_n$ + fertility is conserved

$\Rightarrow$ I construct $\rho^{2^{n-k}}$ rays of $\mathbb{SC}_n$, with $\rho = \#(\text{fertile family of } \mathbb{SC}_k)$

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Theorem (Loho–Padrol–P., '25+)

For $n \geq 5$, *the number of rays of* $\mathbb{SC}_n$ *is bigger than*

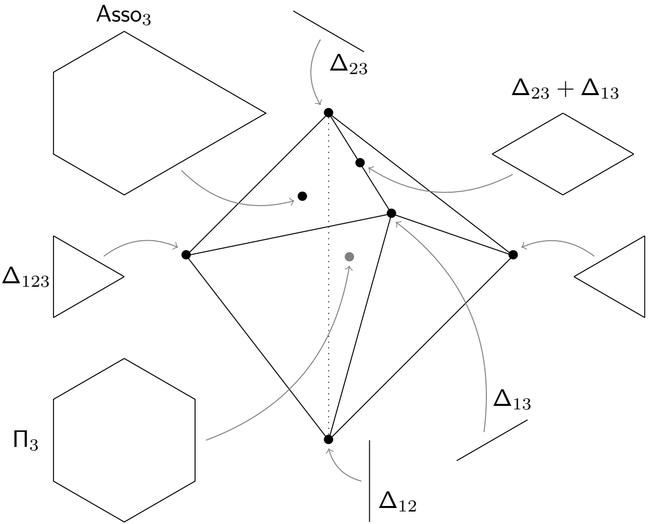
$$656^{2^{n-5}}$$

Going even further beyond

With this induction on (the face lattice of) $\mathbb{SC}_n$:

- Lower bound on the f-vector of $\mathbb{SC}_n$
- Lower bound on the number of non-simplicial faces of $\mathbb{SC}_n$
- New partition of $\mathbb{SC}_n$
- ...

Thank you!

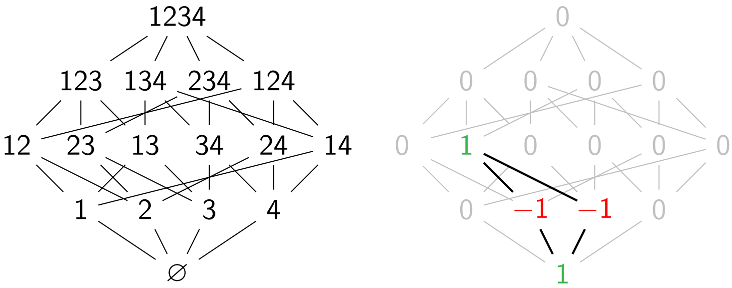


Asso_3
Δ_{23}
$\Delta_{23} + \Delta_{13}$
Δ_{123}
Δ_{13}
Π_3
Δ_{12}

Notations: $Sx = S \cup \{x\}$, $(\boldsymbol{f}_X)_{X \subseteq [n]}$ canonical basis of $\mathbb{R}^{2^{[n]}}$

Definition

Submodular vector $\boldsymbol{n}(S, u, v) = \boldsymbol{f}_{Suv} - \boldsymbol{f}_{Su} - \boldsymbol{f}_{Sv} + \boldsymbol{f}_S$
for $u, v \in S \subseteq [n]$



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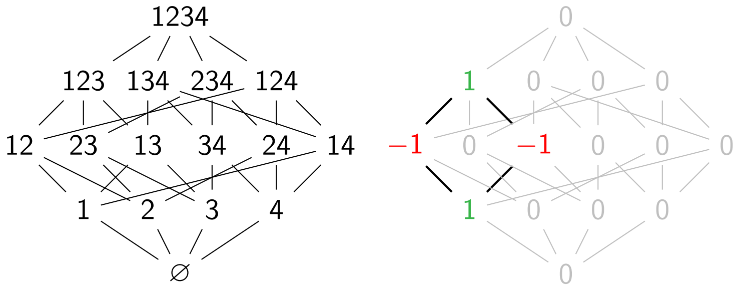

Lemma (Submodular normal)

$\boldsymbol{n}(S, u, v)$ *are the facet's normals of* $\mathbb{DC}(\Pi_n)$

Lemma (Cubic relation)

$u, v, x \notin S \subseteq [n]$

$$\boldsymbol{n}(Suvx, u, v) + \boldsymbol{n}(Sux, u, x) = \boldsymbol{n}(Suv, u, v) + \boldsymbol{n}(Suvx, u, x)$$

$$\begin{tikzcd}
 & Svx \arrow[rr, dash] \arrow[dd, dash] & & Suvx \arrow[dd, dash] \\
Sx \arrow[ur, dash] \arrow[rr, dash] \arrow[dd, dash] & & Sux \arrow[ur, dash] \arrow[dd, dash] & \\
 & Sv \arrow[rr, dash] & & Suv \\
S \arrow[ur, dash] \arrow[rr, dash] & & Su \arrow[ur, dash] &
\end{tikzcd}$$

NB: Cubic relations generates all relations of submodular vectors

Tools: submodular dependancies

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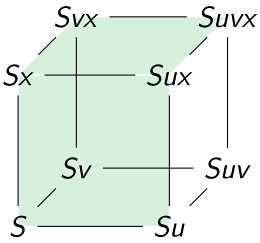

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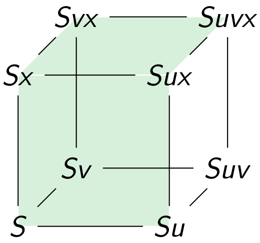

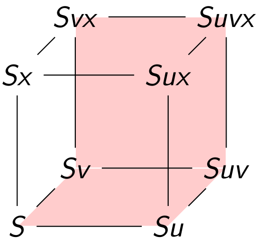

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Fertility

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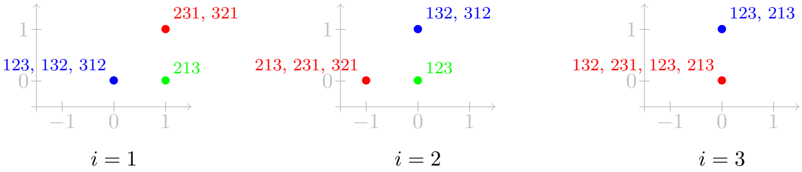

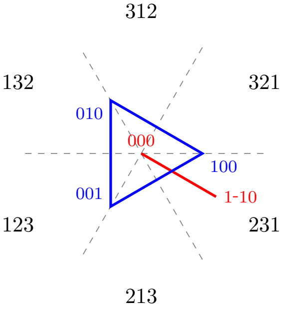

Multiple choice questions
(sometimes several answer possible)

A sum of 2 co-planar triangles can have	3 edges	4 edges	5 edges	6 edges
$\mathcal{N}_{\mathrm{P+Q}} =$ ___ of $\mathcal{N}_{\mathrm{P}}, \mathcal{N}_{\mathrm{Q}}$	union	intersection	common refinement	product
Edge directions of GP $\in$	$\{\boldsymbol{e}_i\}_i$	$\{\boldsymbol{e}_i + \boldsymbol{e}_j\}_{i,j}$	$\{\boldsymbol{e}_i - \boldsymbol{e}_j\}_{i,j}$	$\{\boldsymbol{e}_i \pm \boldsymbol{e}_j\}_{i,j}$
Z_G has no hexagonal face iff G is	a tree	K_3-free	K_4-free	complete
___ form a basis of $\mathrm{Vect}(\mathbb{SC}_n)$	$\{\Delta_X\}_{X\subseteq[n]}$	nestohedra	shard polytopes	matroid polytopes
$\dim \mathbb{SC}_3 =$	3	4	5	11
Group of symmetries of $\mathbb{SC}_n$:	Free_n	$\mathcal{S}_n$	$\mathcal{S}_n \times \mathbb{Z}_2$	$\mathcal{S}_n \times \mathbb{Z}_n$
$\#$ 3-dim indecomposable GP:	3	5	7	37

Exercises

1. Show that $\mathrm{Vert}(\mathsf{P}+\mathsf{Q}) \subseteq \{\boldsymbol{u}+\boldsymbol{v} \;;\; \boldsymbol{u}\in\mathrm{Vert}(\mathsf{P}),\; \boldsymbol{v}\in\mathrm{Vert}(\mathsf{Q})\}$. Find an example with strict inclusion. Prove that $\mathcal{N}_{\mathsf{P}+\mathsf{Q}}$ is the common refinement of $\mathcal{N}_{\mathsf{P}}$ and $\mathcal{N}_{\mathsf{Q}}$.

2. Find all the ways to write the 2-dimensional permutahedron as a Minkowski sum of indecomposable polytopes.

3. Prove that $\boldsymbol{h}_{\mathsf{P}+\mathsf{Q}} = \boldsymbol{h}_{\mathsf{P}} + \boldsymbol{h}_{\mathsf{Q}}$ and $\boldsymbol{\ell}_{\mathsf{P}+\mathsf{Q}} = \boldsymbol{\ell}_{\mathsf{P}} + \boldsymbol{\ell}_{\mathsf{Q}}$. What are $\boldsymbol{h}_{\mathsf{P}+\boldsymbol{t}}$ and $\boldsymbol{\ell}_{\mathsf{P}+\boldsymbol{t}}$?

4. Prove that a triangle is Minkowski indecomposable. Deduce that all simplicial polytopes are Minkowski indecompsable.

Exercises

5. Take vectors $\boldsymbol{s}, \boldsymbol{s}', \boldsymbol{r}_1, \ldots, \boldsymbol{r}_{d-1}$ such that $\mathrm{C} = \mathrm{cone}(\boldsymbol{s}, \boldsymbol{r}_1, \ldots, \boldsymbol{r}_{d-1})$ and $\mathrm{C}' = \mathrm{cone}(\boldsymbol{s}', \boldsymbol{r}_1, \ldots, \boldsymbol{r}_{d-1})$ are simplicial cones which intersect on their proper face $\mathrm{C} \cap \mathrm{C}' = \mathrm{cone}(\boldsymbol{r}_1, \ldots, \boldsymbol{r}_{d-1})$. Show that there exist $\alpha_{\boldsymbol{s}}, \alpha_{\boldsymbol{s}'} > 0$ and $\alpha_{\boldsymbol{r}_i} \in \mathbb{R}$ such that:

$$\alpha_{\boldsymbol{s}} \boldsymbol{s} + \alpha_{\boldsymbol{s}'} \boldsymbol{s}' + \sum_i \alpha_{\boldsymbol{r}_i} \boldsymbol{r}_i = \boldsymbol{0}$$

6. Compute (and draw) the deformation cone of a parallelogram (use the edge-lengths point of view). What about a parallelopiped? Show that there is a **unique** way to write a parallelopiped P as a sum of Minkowski indecomposable polytopes.

7. Show that $\mathbb{SC}_n$ has $\binom{n}{2}2^{n-2}$ facets. Give bounds on its number of rays.

Exercises

8. Show that graphical zonotopes are generalized permutahedra. Show that nestohedra are generalized permutahedra. Show that matroid polytopes are generalized permutahedra. Which are indecomposable?

9. If G is triangle-free, then $\mathbb{DC}(\mathrm{Z}_G)$ is simplicial (see lecture). Each face of $\mathbb{DC}(\mathrm{Z}_G)$ is associated to a polytope: which polytope?

10. The *weighted graphical zonotope* of a graph $G = (V, E)$ and weight (on edges) $\omega : E \to \mathbb{R}_+^*$ is $\mathrm{Z}_{G,\omega} := \sum_{(i,j)\in E} \omega(i,j)[\boldsymbol{e}_i, \boldsymbol{e}_j]$. Show that: $\{\text{zonotopes}\} \cap \{\text{generalized permutahedra}\} = \{\text{weighted graphical zonotopes}\}$. Show that the cone of weighted graphical zonotopes is a section (i.e. intersection with a linear sub-space) of the submodular cone, such that the rays of this section are rays of $\mathbb{SC}_n$. What is the dimension of this section?

11. Prove the cubic relations hold. Show that cubic relations generates all linear relations between submodular vectors.